الأسئلة المقالية — 10 متقدم · مذاكرة الهيكل الاختبار النهائي للفصل الاول

الاسم: _____ المدرسة: _____ التاريخ: _____ (20 سؤالاً)

LO21 التحليل باستعمال المتطابقات الخاصة · Special-product factorization

21.1 LO21 · Reveal p. 680 · 35

Factor completely using integers.

$$-72p^2 + 8x^2$$

a. Show your factorisation and check by expansion. (10)

الأستاذ كامل بشارة 0502837825

21.2 LO21 · Reveal p. 680 · 38

Factor completely using integers.

$$16b^2 - 100$$

a. Show your factorisation and check by expansion. (10)

الأستاذ كامل بشارة 0502837825

21.3 LO21 · Reveal p. 681 · 51

Determine whether this is a perfect-square trinomial; if so, factor it.

$$9z^2 - 6z + 1$$

a. Show your factorisation and check by expansion. (10)

21.4 LO21 · Reveal p. 681 · 52

Determine whether this is a perfect-square trinomial; if so, factor it.

$$36x^2 - 60x + 25$$

a. Show your factorisation and check by expansion. (10)

LO22 الاستدلال بالزوايا الداخلية والخارجية · Interior and exterior angle reasoning

22.1 LO22 · Reveal p. 64 · 9

The given angle is an interior angle of a regular polygon. Find the number of sides.

144°

a. Show an equation and solve for n. (10)

22.2 LO22 · Reveal p. 64 · 11

The given angle is an interior angle of a regular polygon. Find the number of sides.

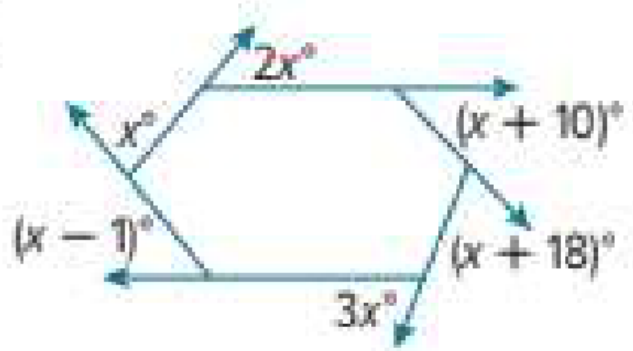
160°

a. Show an equation and solve for n. (10)

22.3 LO22 · Reveal p. 64 · 15

Find x in the diagram.

15.

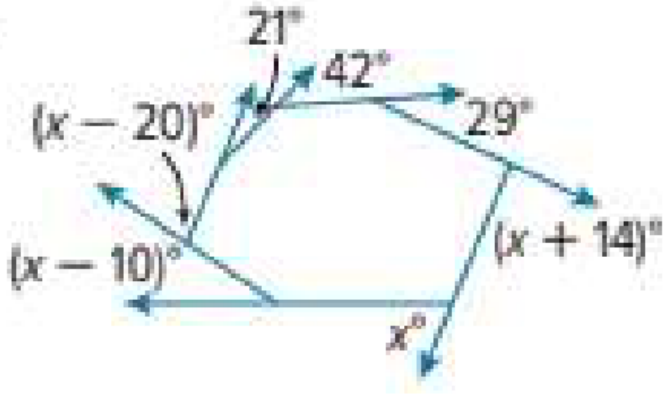


a. Show how you use the exterior-angle sum. (10)

22.4 LO22 · Reveal p. 64 · 18

Find x in the diagram.

18.

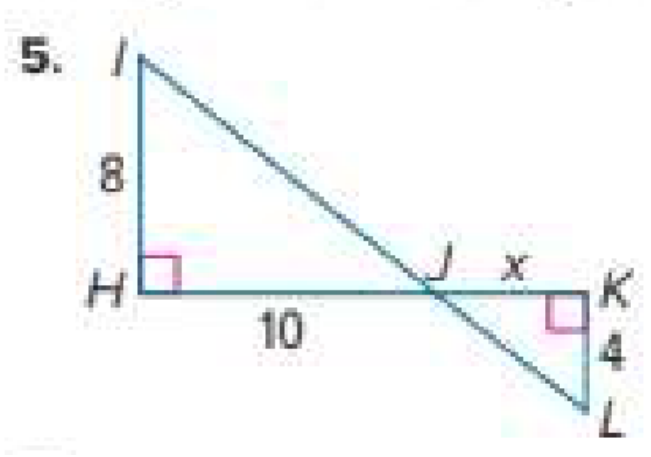


a. Show how you use the exterior-angle sum. (10)

LO23 حل المسائل باستعمال المثلثات المتشابهة · Solving with similar triangles

23.1 LO23 · Reveal p. 139 · 5

Identify the similar triangles. Then find x.

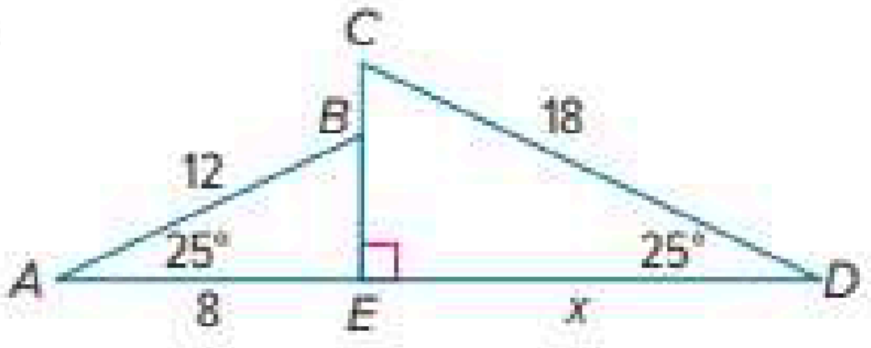


a. State corresponding vertices and show a proportion. (10)

23.2 LO23 · Reveal p. 139 · 6

Identify the similar triangles. Then find x.

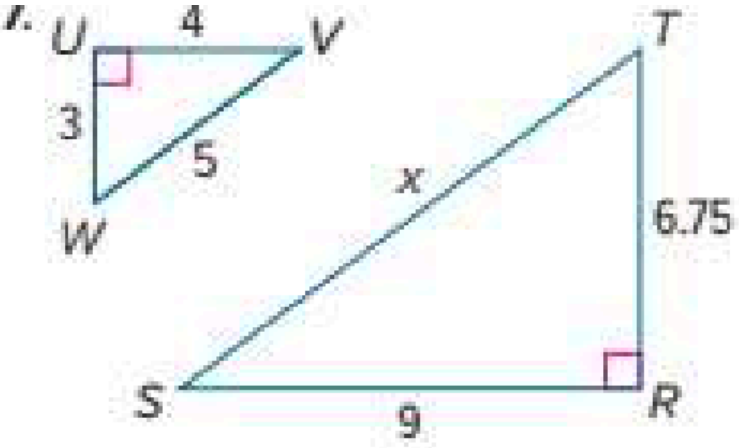
6.



a. State corresponding vertices and show a proportion. (10)

23.3 LO23 · Reveal p. 139 · 7

Identify the similar triangles. Then find x.

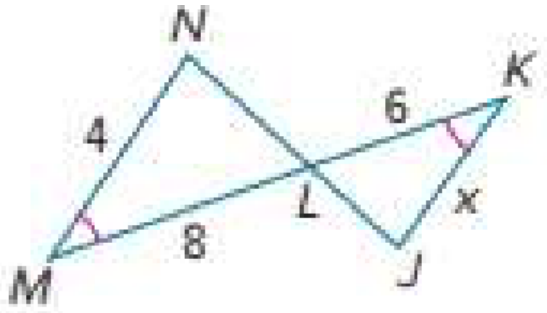


a. State corresponding vertices and show a proportion. (10)

23.4 LO23 · Reveal p. 139 · 8

Identify the similar triangles. Then find x.

3.



a. State corresponding vertices and show a proportion. (10)

LO24 المسافة ومنتصف القطعة في الفراغ · Distance and midpoint in space

24.1 LO24 · Reveal p. 177 · 13

Find the midpoint of the segment joining the two points.

$$K(-2,-4,-4), \quad L(4,2,0)$$

a. Determine the coordinates of midpoint M. (10)

24.2 LO24 · Reveal p. 177 · 15

Find the midpoint of the segment joining the two points.

$$R(3, 3, 4), \quad V(5, 4, 13)$$

a. Determine the coordinates of midpoint M. (10)

24.3 LO24 · Reveal p. 178 · 19

Find the distance and midpoint of the segment.

$$P(-5, -2, -1), \quad Q(-1, 0, 3)$$

a. Determine the distance. (5)

b. Determine the coordinates of midpoint M. (5)

24.4 LO24 · Reveal p. 178 · 23

Find the distance and midpoint of the segment.

$B(\sqrt{3}, 2, 2\sqrt{2}), \quad C(-2\sqrt{3}, 4, 4\sqrt{2})$

a. Determine the distance. (5)

b. Determine the coordinates of midpoint M. (5)

25.1 LO25 · Reveal p. 208 · 15

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 50^\circ, \quad a = 34, \quad b = 40$$

a. Determine the number of solutions and justify it. (4)

b. Give every possible triangle, or explain why none exists. (6)

25.2 LO25 · Reveal p. 208 · 16

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 24^\circ, \quad a = 3, \quad b = 8$$

a. Determine the number of solutions and justify it. (4)

b. Give every possible triangle, or explain why none exists. (6)

25.3 LO25 · Reveal p. 208 · 17

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 125^\circ, \quad a = 22, \quad b = 15$$

a. Determine the number of solutions and justify it. (4)

b. Give every possible triangle, or explain why none exists. (6)

25.4 LO25 · Reveal p. 208 · 19

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 30^\circ, \quad a = 2, \quad b = 4$$

a. Determine the number of solutions and justify it. (4)

b. Give every possible triangle, or explain why none exists. (6)